

Carbon Clarity: A Transparency and Accountability System for Community Carbon Finance.

Prepared for the **International Organization for Migration (IOM) Mozambique**.
Nine resettlement sites in **Cabo Delgado, Nampula, and Niassa**, Mozambique.
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EXECUTIVE SUMMARY

Displaced communities in northern Mozambique are being asked to trust a carbon market they cannot see, understand, or verify. In 2024, IOM Mozambique distributed 2,770 improved cookstoves across nine resettlement sites, expecting carbon credits to generate income for households, but the carbon finance component failed, credits were not registered, no payments arrived, and no transparent explanation was given. In a region where many women have limited formal education and literacy remains low; this deepened an existing transparency and trust gap between vulnerable households and the institutions managing climate finance in their name. The core problem is **information asymmetry**: communities generate climate value but cannot see the market price, the costs deducted, or the formula that determines their share, and research on African voluntary carbon markets shows that local communities often receive only a small fraction of total revenues under opaque arrangements.

Carbon Clarity addresses this problem with a **two-part transparency system** that works inside existing IOM and developer structures. First, the **Carbon Clarity Hub** uses trusted channels, **Village Savings and Loan Associations (VSLAs)**, visual household guides in **Emakhuwa**, and **community radio**, to build carbon literacy over 12 months through short, recurring sessions, instead of one-off workshops. **Public price boards** in village centers translate abstract carbon prices into simple visual steps showing market price, real costs, community share, and household payment, so people can follow how value flows from "**what the world pays**" to "**what you receive**." Second, the **Transparent Intermediary** focuses on enforcement: every payment comes with an **itemized**, **icon-based receipt**; **elected Carbon** Oversight Committees with **reserved seats** for women, youth, and people with disabilities review pricing and community funds; VSLA Carbon Liaisons help neighbors read receipts and file complaints; and all grievances are handled through IOM's existing Complaints and Feedback Mechanism with a 14-day resolution target.

Pilot testing with 50 household representatives across three sites suggests this low-tech, visual-first approach is both understandable and acceptable: when a text-heavy price board was replaced with a simplified **icon-based** version, comprehension improved significantly, and communities expressed a preference for grievance routes that run through familiar local actors and

IOM's systems rather than distant hotlines. By the end of Year 3, the project aims for at least 80 percent of households to be able to explain how their carbon payment is calculated, for 100 percent of payments to be accompanied by receipts that can be checked against **public price boards**, and for 2,770 households to receive transparent, verifiable carbon income, alongside community benefit funds financing priorities like water, education, or health. For IOM, this reduces reputational and safeguarding risks while turning carbon finance into a credible recovery tool; for developers and buyers, it offers a concrete way to meet emerging requirements from Gold Standard, Verra, and ICVCM on community engagement, transparency, and grievance mechanisms, supporting more trustworthy, socially grounded carbon credits.

INTRODUCTION

This project began with a simple but difficult question: how can displaced communities in northern Mozambique understand, verify, and trust carbon market transactions when the system is distant, technical, and slow, and when they are already living with insecurity and scarcity? Since 2017, conflict in Cabo Delgado and neighboring provinces has forced hundreds of thousands of people from their homes, pushing families into resettlement sites where access to energy, income, and information is extremely limited. Most displaced households rely on biomass for cooking, women and girls spend hours each day collecting firewood, and the country as a whole continues to lose large areas of forest each year to fuel, agriculture, and other pressures. In this context, improved cookstoves and carbon finance are attractive because they promise both environmental and livelihood benefits, but only if communities can see and influence how value is created and shared.

The 2024 IOM cookstove initiative highlighted both the potential and the risk of relying on carbon markets in displacement settings. On the one hand, distributing 2,770 stoves across nine sites reduced fuel use and exposure to smoke; on the other hand, the carbon-credit component failed to deliver any income, and households received no clear, accessible explanation of what went wrong. This exposed a deeper structural issue: contracts and benefit-sharing formulas were written in Portuguese or English, communities were never taught how carbon pricing and verification work, and there were no simple tools to check whether promised shares matched actual payments. In a setting where female illiteracy is high and many adults have limited formal schooling; this information gap is not a minor inconvenience: it effectively excludes the very people whose daily cooking practices generate the carbon value being sold.

For stakeholders such as IOM, carbon project developers, and carbon-credit buyers, this challenge is more than a communication problem; it is a question of legitimacy and risk. If

communities cannot understand or verify benefit-sharing, they cannot give meaningful consent, and any breakdown, such as the 2024 failure, undermines trust in future projects and creates reputational and safeguarding risks for agencies on the ground. At the same time, forests in northern and central Mozambique continue to be cleared, and clean cooking access remains low, meaning there is a genuine opportunity for well-designed carbon projects to support both climate and development goals. The project therefore defined its boundaries narrowly but deliberately: it does not try to change global carbon markets or replace developers, and it does not duplicate IOM's operational role in distribution or monitoring. Instead, it focuses on the community-facing layer of the system, education, transparency tools, governance structures, and grievance mechanisms, for the 2,770 households across nine IOM cookstove sites. By tackling information asymmetry and power imbalance at this level, the design aims to protect communities from unfair deals, help IOM meet its protection and accountability commitments, and give buyers greater confidence that climate finance is reaching the people whose behavior change makes the credits possible.

DESIGN PROCESS

The design process started from a simple observation: communities were being asked to trust a carbon system they did not understand and could not verify, in a context where displacement, poverty, and low literacy already limited their power. To respond responsibly, the team first had to understand people's lived experiences of cookstoves, income promises, and failed projects, and then translate those insights into a concept that was both ambitious and realistic for IOM's operational constraints.

Empathize & Define: What We Learned from Users

The empathy work drew on three main sources: conversations with IOM Mozambique staff, review of the 2024 cookstove experience, and secondary research on clean cooking and displacement. IOM staff described how women in Cabo Delgado, Nampula, and Niassa spend hours each day collecting fuelwood and cooking over smoky fires, often in crowded shelters that expose children to high levels of indoor air pollution. Studies on clean cooking in displacement settings confirm that, when cleaner technologies are introduced without adequate support, adoption drops off over time if people do not see clear, sustained benefits or if they cannot afford or trust the new solutions.

The 2024 stove initiative added a specific carbon finance shock. While the stoves themselves delivered real gains, the promised carbon payments never materialized. From the community's perspective, this looked like yet another "project" where outsiders made commitments in technical language, took data and photos, and then left without clear accountability. At the same time, IOM and other stakeholders were under pressure to show results on climate and energy, in line with global calls to integrate clean cooking into basic service packages for displaced populations. The result was a shared frustration: communities felt misled,

and institutional partners worried that one bad experience might poison the well for future climate-finance opportunities.

From these insights, the team articulated a refined problem statement. The issue was not simply "communities do not know what carbon is," but rather: displaced households in northern Mozambique generate measurable climate value through their daily use of improved cookstoves, yet they are structurally unable to see, verify, or negotiate how that value is priced and distributed. Contracts, cost deductions, and benefit-sharing formulas remain invisible and incomprehensible at community level, while information flows one-way from developers to registries. This opacity enables extractive outcomes, erodes trust, and directly undermines long-term stove adoption and climate impact. The design challenge, therefore, is not awareness alone, but closing the information and power gap without attempting to control carbon markets or registry processes.

Ideate: Exploring Options and Choosing a Direction

With the problem better defined, the team generated several possible solution pathways, ranging from high-tech dashboards to community cooperatives. One idea was a mobile or blockchain-based system where each household could see real-time estimates of the carbon they had generated and the revenue due to them. While attractive on paper, this approach was quickly judged unrealistic in a setting where electricity is unreliable, smartphone penetration is low, and many users struggle with text-based interfaces. Another idea was to create a community carbon cooperative that would aggregate credits and sell directly to buyers, cutting out middlemen. However, discussions with IOM leadership (staff) and reference to sector guidance on humanitarian innovation showed that this would require a new, complex organizational structure and revenue model, with high risk and little guarantee that developers or buyers would participate.

A third option focused only on simplifying contracts and translating them into Emakhuwa. While translation is important, experience from other projects and grievance mechanism guidance suggested that a short contract alone does not build understanding or enforce accountability; people must be able to see how principles play out in concrete numbers, at the moment payments are made, and must have somewhere to go if those agreements are not respected. Evidence from clean cooking and payment-for-ecosystem-services projects also emphasizes that ongoing, two-way communication and accessible complaint channels matter more than one-off consultations.

The concept that best balanced ambition and feasibility was a low-tech, community-based transparency system built on platforms that already work in displacement settings: VSLAs, community radio, and IOM's existing Complaints and Feedback Mechanism. This approach scored highly on feasibility (no reliance on new digital infrastructure), user impact (directly targets understanding and verification), cost-effectiveness (can operate on a modest budget), and sustainability (can be embedded into IOM's routine operations once piloted). By choosing this direction, the team accepted that the solution would not control carbon prices or registry timelines, but would instead give communities clear tools to understand, track, and challenge how carbon revenues are shared.

Prototype & Test: Making It Real and Improving It

Prototyping focused on a small set of tangible tools: **visual household guides**, **public price boards**, **itemized receipts**, **VSLA training modules**, and a **grievance workflow** that plugs into **IOM's systems**. Early versions of the guides and boards were drafted with more text and numeric detail, under the assumption that "more information is better." When tested with around 50 household representatives in three pilot sites through participatory workshops, this assumption proved wrong. Participants found the text-heavy designs confusing, and only a minority could correctly explain how a sample payment had been calculated. After redesigning the boards and receipts to use clear icon sequences and minimal text in Emakhuwa, comprehension improved substantially, with many participants able to retell the logic to others.

Similar lessons emerged from testing training formats and grievance options. Intensive, one-day training left people overwhelmed and unable to recall key points a few weeks later, whereas short, monthly sessions linked to existing VSLA meetings were easier to absorb and sustain. Participants also expressed discomfort with remote, anonymous complaint lines, preferring to raise concerns through trusted local figures who could then connect to formal systems. In response, the design shifted to: a 12-month VSLA-based learning sequence, highly visual materials, price boards and receipts that mirror each other, and a grievance pathway that begins with VSLA Carbon Liaisons and flows into IOM's established feedback mechanism with clear timelines. These iterations grounded the final concept in users' real preferences and constraints, while aligning it with good practice from humanitarian innovation and grievance mechanism literature.

FINAL SOLUTION

Carbon Clarity is a low-tech transparency system that makes carbon finance understandable, verifiable, and contestable for displaced communities. Rather than replacing existing actors, it embeds education, verification, and accountability mechanisms into trusted institutions already operating in displacement settings. By doing so, it transforms carbon finance from an opaque, expert-driven process into a system communities can follow, question, and influence. The first part builds understanding and expectations using channels communities already trust, such as VSLAs, visual guides, and radio. The second part makes every payment traceable and contestable through receipts, public price boards, community oversight, and an integrated grievance pathway that runs through IOM's Complaints and Feedback Mechanism rather than a parallel structure. Together, these elements turn an invisible, expert-driven system into something that a woman in a resettlement site can follow, question, and influence.

On the **educational side**, the Carbon Clarity Hub is built around a 12-month curriculum delivered through existing VSLA meetings. Instead of overwhelming people with a single intensive training, the model uses short, structured sessions that gradually introduce key ideas: what carbon credits are, why payments are delayed, how market prices work, which costs are legitimate, and what share should reach the community. Visual household guides in Emakhuwa,

with icons and minimal text, reinforce these lessons at home. Community radio episodes, written as stories rather than lectures, dramatize common situations, such as a woman checking her receipt against the price board and discovering an error, and show how to resolve them. Public price boards installed in central locations then act as the bridge between education and practice: they show, in a simple step-by-step flow, the current market price per tonne, the costs deducted, the percentage allocated to the community, and the final payment per household. This combination of repeated, oral explanations and visual references is designed to work in low-literacy, high-stress displacement environments where time and attention are scarce.

On the enforcement side, the **Transparent Intermediary** translates abstract fairness principles into concrete safeguards at each payment cycle. Every household receives an itemized receipt with icons that match the price board, so they can check whether the agreed formula has been applied correctly. If the board says that a family with two tonnes of emissions reductions should receive a specific amount after costs and community contributions, the receipt must show the same logic. Elected Carbon Oversight Committees, with seats reserved for women, youth, and persons with disabilities, review the cost assumptions, confirm that payments match the formula, and oversee the use of the community benefit fund. VSLA Carbon Liaisons act as local interpreters and advocates, helping neighbors understand their documents and logging complaints into IOM's existing CFM, which already follows international guidance on feedback and grievance redress. This design reflects lessons from benefit-sharing and grievance-mechanism literature: people are more likely to use and trust complaint systems when they can access them through known intermediaries and when they see that issues are resolved within clear timeframes.

Operationally, **Carbon Clarity** is intentionally low-tech and light-footprint. The core materials are laminated guides, hard-wearing price boards, standardized receipt pads, and simple training scripts, all of which can be produced locally and maintained at modest cost. The human "infrastructure" relies on roles that already exist or can be added with small stipends: VSLA facilitators delivering sessions, liaisons supporting verification and grievance logging, and IOM field staff integrating carbon issues into their routine community engagement. This means the system can function even with limited electricity and connectivity, and it aligns with global recommendations that clean cooking and energy interventions in displacement settings should be designed for constrained environments, not ideal ones. At the same time, the model is compatible with emerging high-integrity cookstove methodologies and buyer expectations: by making benefit-sharing formulas explicit and auditable, it supports claims that credits from these projects meet stronger transparency and community-engagement standards.

In practice, the solution resolves the key pain points identified during research. The information gap is addressed by breaking down complex concepts into repeated, visual explanations that are anchored in daily life and familiar institutions. The powerlessness communities described is addressed by giving them concrete tools, price boards, receipts, committees, and grievance channels, to check and contest payment decisions. The trust deficit is tackled not by asking people to believe promises, but by letting them see, step by step, how money flows and how mistakes or abuses can be corrected. Because the design is modular and based on

widely used structures like VSLAs and CFMs, it can also adapt to other clean cooking or carbon projects and be scaled to new sites without rebuilding the system from scratch. In short, Carbon Clarity is a technically modest but strategically powerful intervention: it makes the carbon market legible and negotiable for displaced households, while helping institutions like IOM, developers, and buyers meet their own standards for accountability, benefit sharing, and climate impact.

USER FEEDBACK AND REFINEMENT

The solution was not designed on paper alone; it was shaped by how real users reacted to early versions of the tools and processes. To understand whether Carbon Clarity would actually work for displaced households, the team organized small-scale testing with around 50 household representatives across three sites, using participatory workshops and focus group-style discussions to observe how people interacted with visual guides, price board prototypes, receipt templates, and draft training sessions. This approach is consistent with good practice in the clean cooking and humanitarian sectors, which emphasize direct user trials and iterative refinement rather than assuming that technically correct designs will automatically be adopted.

During these sessions, participants were first shown more traditional, text-heavy materials, a price board listing the market price per ton, a breakdown of transport and verification costs, and a numerical formula for calculating the community share and household payments. Facilitators then asked simple questions such as, “How much should Amina receive if she generated two tons?” and “Can you explain to your neighbor how this number was calculated?” In this round, only a minority could follow the logic from top to bottom, and many said the information looked like “office work” rather than something meant for them. When the team replaced the text and dense numbers with an icon sequence showing the world price, the portion allocated to the community, the costs deducted, and the final payment per household, comprehension improved sharply. Participants began to narrate the steps themselves, and several volunteered to explain the board back to the group. This mirrors broader market research findings: visual and story-based communication tends to outperform purely textual messages in low-literacy contexts.

Feedback on training formats led to similarly important refinement. The original idea of a one-day intensive workshop on carbon markets, benefit-sharing, and grievance procedures seemed efficient from a project-management perspective, but pilot participants reported feeling overwhelmed, and follow-up checks showed that many could not recall key concepts a few weeks

later. In response, the curriculum was broken into short, monthly sessions embedded in existing VSLA meetings, with each session focusing on a single topic reinforced through repetition, questions, and simple exercises. Evidence from clean cooking and behavior-change programs supports this shift: long-term adoption is more likely when information is delivered in manageable pieces over time and tied to familiar routines rather than isolated events.

The design of the grievance pathway was also adjusted based on user preferences and lessons from feedback-mechanism literature. Early drafts considered a standalone hotline dedicated to carbon complaints, but participants expressed discomfort with calling unknown numbers and fear that complaining directly to a developer might jeopardize future benefits. They said they would rather talk to someone they know and trust, such as a VSLA leader or an IOM field worker, and have that person help them "take the issue forward." This aligns with wider experience in humanitarian projects, which shows that community-based access points and integration into existing complaint and feedback mechanisms increase use and trust. The final model therefore routes grievances through VSLA Carbon Liaisons and Oversight Committees, who record cases and enter them into IOM's Complaints and Feedback Mechanism, where standard procedures for follow-up and resolution already exist.

User feedback also highlighted the importance of representation and voice, not just information. Women participants in particular raised concerns that, even if they understood the price board and receipts, decisions about community funds or negotiations with developers might still be dominated by male leaders or external actors. In response, the governance design was refined to include explicit quotas for women, youth, and people with disabilities in Carbon Oversight Committees, and to require that major fund allocations be discussed in open community meetings. This is in line with evidence from VSLA and community-based governance initiatives, which shows that formal rules and quotas, combined with financial transparency, can increase the meaningful participation of marginalized groups in decision-making.

These refinements did more than improve user comprehension; they changed the tone of conversations about carbon finance. In later workshops, participants began to ask more specific, rights-based questions, such as what would happen if market prices changed, or how they could know whether reported costs were reasonable, indicating a shift from passive acceptance to active

scrutiny. By grounding each design adjustment in observed behavior and feedback, and by aligning those changes with broader evidence on clean cooking adoption, grievance mechanisms, and VSLA governance, the project increased the likelihood that Carbon Clarity will be both usable for communities and credible for institutional stakeholders.

IMPACT ASSESSMENT

Carbon Clarity delivers tangible improvements across four dimensions: social agency and trust, environmental outcomes, economic benefits, and institutional credibility. These impacts are interconnected; together, they show how transparency can transform carbon finance from a source of disappointment into a real tool for both climate action and community resilience in displacement settings.

Social Impact: Restoring Agency and Trust

The most direct measure of success is whether communities understand and can verify their carbon payments. The baseline is stark: after the 2024 failed pilot, only around 10 percent of households could explain what a carbon credit is or how benefit-sharing works. By the end of Year 3, the target is for 80 percent to be able to explain, in simple terms, how their payment is calculated and to verify it against a public price board. This shift from powerlessness to verification is not merely educational; it is a fundamental change in the power dynamic between households and the system managing their climate finance. Research on grievance mechanisms and benefit-sharing in development projects shows that, when people can see and challenge unfair arrangements, they are more likely to trust the system and participate in future initiatives. For displaced communities in Mozambique who have experienced broken promises around energy, income, and protection, restoring this trust is essential for long-term adoption of clean cooking and engagement with climate finance.

A second social outcome is women's leadership and voice. Baseline data shows that women in most resettlement sites attend VSLA meetings but rarely hold formal decision-making roles. Carbon Clarity's design builds in explicit quotas: at least 3 of the 7 seats on each Benefit Fund Committee must be held by women, and at least 50 percent of VSLA Carbon Liaisons should be women. By the end of Year 3, this translates to approximately 50 women in formal governance positions across nine sites, making decisions about carbon pricing, community fund use, and grievance resolution. Evidence from VSLA and community-based resource management programmes shows that when women hold formal roles with clear authority and resource access, household-level benefits often increase, women tend to prioritize spending on education, health, and nutrition, and community cohesion improves. By embedding women's leadership into carbon governance, the project creates a secondary benefit: it models and strengthens women's voice in other community decisions.

Environmental Impact: Sustained Behavior Change and Forest Protection

Improved cookstoves deliver health and time-saving benefits immediately, but long-term adoption depends on whether households see sustained value. In projects without transparent benefit-sharing, stove usage often declines over time as people revert to charcoal or firewood if they feel cheated or see no real income gain. The target for Carbon Clarity is to maintain an 85 percent stove retention rate by Year 3, compared to a typical 70 percent baseline in similar projects. This matters for forests: each stove saves approximately 2.5 to 3 tonnes of biomass per household annually. Across 2,770 households, sustained use means avoiding the cutting and burning of roughly 8,000 tonnes of fuelwood per year. Over a 10-year horizon, this translates to protection of thousands of hectares from conversion to charcoal production and other uses.

Beyond fuel savings, better stove adoption also delivers measurable health gains. Indoor air pollution from cooking fires is a leading cause of respiratory disease and respiratory complications in pregnancy in sub-Saharan Africa. Women and children in households that consistently use improved stoves report reduced coughing, fewer respiratory infections, and less eye irritation. While Carbon Clarity does not directly provide health services, by making stove use more sustainable through transparent carbon income and community benefit investments (such as health worker stipends or water points), the project indirectly supports health outcomes. This alignment with broader development goals, clean energy, health, and gender equality, means that Carbon Clarity is not a single-issue climate intervention; it is a vehicle for multiple, interconnected improvements.

Economic Impact: Reliable Income and Community Priorities

At the household level, carbon income provides a meaningful but not transformative addition to family budgets. The target is for each participating household to receive approximately $12 to $15 annually in verified carbon payments by Year 3. In a displacement context where informal labor income averages $1 to $3 per day, this represents the equivalent of 1 to 2 weeks of income, enough to matter for school fees, health costs, or emergency savings without creating dependency. The key advantage is reliability and visibility: unlike irregular informal labor, carbon payments arrive on a predictable schedule with a documented, transparent explanation. This allows households to plan, and it reinforces the message that communities are valued contributors to climate action rather than passive beneficiaries of external projects.

At the community level, carbon revenue also funds collective priorities. The design allocates a small percentage (e.g., 5 percent) of carbon payments to a community benefit fund, managed by the Oversight Committee. Across nine sites, this generates approximately 10,000 Meticais per site, enough to fund a water point repair, school furniture, community health worker stipends, or emergency savings, with the community choosing the priority. This approach, rooted in participatory governance literature, ensures that climate finance benefits are visible and locally controlled, rather than allocated by external actors. It also creates a second tier of impact: when communities use carbon money to invest in water, health, or education, they are directly linking climate action to development outcomes that matter most to them.

Systemic Impact: Market Credibility and Standards Alignment

Carbon Clarity also benefits the broader system. Developers and carbon buyers face increasing pressure from standards bodies (Gold Standard, Verra, the ICVCM Core Carbon Principles) to demonstrate genuine community engagement, transparent benefit-sharing, and functioning grievance mechanisms. Many projects struggle to meet these expectations at scale because they lack simple, proven tools. By demonstrating that a low-cost, VSLA-based transparency model can reliably deliver community understanding, documented consent, and predictable dispute resolution, Carbon Clarity provides a replicable blueprint that developers can incorporate into their own projects. This potentially unlocks a price premium: carbon buyers increasingly pay a 10 to 20 percent premium for credits backed by strong social integrity and transparent community approval, a "Socially Certified" label that Carbon Clarity helps projects earn. For IOM, it reduces reputational and safeguarding risks and demonstrates that the organization can work with climate finance in ways that strengthen, rather than undermine, community protection.

In sum, Carbon Clarity delivers concrete, measurable improvements across social agencies, environmental stewardship, household economics, and institutional credibility. These impacts are not independent; together, they create a feedback loop in which transparency leads to understanding, understanding builds trust, trust sustains stove use, sustained use protects forests and delivers health gains, visible benefits strengthen community voice, and strong community voice ensures that future climate finance is designed with accountability from the start. This is why Carbon Clarity matters not just for the 2,770 households in nine Mozambique sites, but as a model for how climate finance can serve displaced and vulnerable communities anywhere.

IMPLEMENTATION PLAN (EXCEL SHEET)

Carbon Clarity is not a one-time event but a carefully sequenced 36-month process that builds understanding and systems before payments arrive, then enforces transparency as money flows. This plan utilizes a **$118,000 core budget** for operations, while holding a **$32,000 Risk & Inflation Reserve** to ensure resilience in northern Mozambique. This phasing reflects how carbon projects actually work, there is a lag between stove distribution and credit registration, and it uses that lag strategically to prepare communities and institutions for fair, transparent benefit-sharing.

The phased budget allows for a high-intensity setup in Phase 1 ($97,900), transitioning to a leaner, self-sustaining operating model by Year 3. In Years 1–2, this budget is intentionally supported by grant funding to absorb early-stage risk, establish facilitation capacity, and build trust before carbon revenues are available. By Year 3, the same operating envelope is increasingly financed through service fees, training revenues, and initial carbon margins, reflecting the transition from system setup to market validation.

Phase 1: Foundation & Education (Months 0–18)

The first phase focuses on getting permission, building relationships, and establishing that communities understand carbon concepts before any money arrives. In the first two months, the Carbon Clarity team finalizes a Memorandum of Understanding with IOM Mozambique, confirming partnership, access to VSLA sites, and integration with the Complaints and Feedback Mechanism. At the same time, the carbon project developer must formally commit to transparency, signing a contractual clause that commits to sharing pricing data, cost information, and payment schedules with communities. These two documents are not optional; they are critical-path dependencies. Without them, the project cannot proceed.

By month 6, the team conducts participatory co-design workshops in each of the nine sites, testing visual guides, price boards, and training scripts with community members and incorporating their feedback into revised versions. Eighteen VSLA facilitators (2 per site) receive intensive two-day training on carbon basics, facilitation skills, and how to teach adults in low-literacy contexts. Simultaneously, existing IOM field staff receive a one-day orientation on carbon pricing and grievance issues so they can support the VSLA facilitators and answer basic questions from communities. No new staff are hired; the work relies on stipends for facilitators and training for existing staff.

From months 7 to 18, the five Phase-1 VSLA training modules are delivered monthly at each site, covering: what carbon credits are, why payments take time, non-carbon benefits of cookstoves, how pricing formulas work, and reading the price board. In parallel, 2,770 visual guides in Emakhuwa are printed and distributed to households, community radio content is produced and broadcast, and Carbon Oversight Committees and VSLA Carbon Liaisons are elected at each site. By month 18, a baseline survey of approximately 500 households' measures carbon literacy, trust levels, and understanding of rights. The target is that at least 60 percent of VSLA members can explain carbon concepts, and all governance structures are in place and trained.

Phase 2: Deepening & Negotiation (Months 19–36)

Phase 2 is when communities move from learning to doing. The Phase-2 VSLA curriculum focuses on deeper topics: understanding costs, negotiating benefit-sharing formulas, and practicing grievance procedures. Simultaneously, Carbon Oversight Committees hold monthly meetings to review pricing assumptions and prepare to negotiate with the developer on the final benefit-sharing formula. By month 24, communities and the developer have formally agreed on what percentage of carbon revenue goes to households versus operational costs, and this agreement is documented and public. This is crucial: it means that when credits are eventually issued and payments flow, there is no ambiguity about what is "fair." By months 25–36, the first payments arrive with itemized receipts, price boards are updated with real numbers, and the grievance system springs into active use, with VSLA liaisons logging complaints into IOM's Complaints and Feedback Mechanism and aiming for 14-day resolution.

Phase 3: Payment & Accountability (Months 37+)

Phase 3 begins when credits are issued by the registry and money flows to the developer, who then distributes to communities. Every payment is accompanied by a receipt matching the public price board, community committees overseeing fund use, and any complaints are tracked and resolved. This phase continues indefinitely; carbon projects often produce payments for 7 to 10 years or more. The work becomes routine: facilitators continue monthly VSLA discussions, liaisons help households understand receipts, committees manage community benefit funds, and grievances are handled through established channels.

Resources & Timeline Summary

The total budget for all three phases is approximately $118,000 over three years, or about $16 to $17 per household. This breaks down as: facilitator and liaison stipends ($54,000, the largest component, recognizing local labor); materials such as guides, boards, and receipt pads ($25,000); staffing for a part-time Coordinator and M&E support ($14,400); radio programming ($9,600); and training and operations ($15,000). Year 1 requires the most investment ($45,000) because of one-time setup and training costs. Years 2 and 3 are less intensive ($38,000 and $35,000 respectively) as materials are reprinted and systems are refined. This budget is affordable for climate innovation funders and demonstrates that transparency does not require expensive technology or heavy staffing. Importantly, Carbon Clarity does not rely on indefinite budget growth to achieve sustainability. From Year 2 onward, core operating costs remain broadly stable, while the funding mix shifts from grants to earned revenue. By Years 4–5, the same operating budget is fully covered through developer and institutional service fees, carbon margin revenue, and paid training or replication support, eliminating long-term donor dependency while preserving affordability for community-facing activities.

We assume that Carbon Clarity can **piggyback** on IOM Mozambique's existing logistics, security, and office infrastructure at **zero additional cost**, allowing our budget to focus exclusively on transparency tools and stipends. The core operational budget of $118,000 is bolstered by a $32,000 Risk & Inflation Reserve, bringing the total initial commitment to $150,000.

Roles & Accountability

Clear accountability is essential for a system based on transparency. The Carbon Clarity Coordinator oversees the project and reports to both the funder and IOM. IOM's Programme Director holds operational responsibility and ensures that field staff support VSLA work and integrate carbon issues into safeguarding. VSLA facilitators deliver training and manage peer learning; liaisons handle community intake of concerns; Carbon Oversight Committees make local decisions and represent communities in discussions with developers. The external evaluator conducts baseline, midterm, and endline assessments, providing independent evidence of whether targets are met. Each actor has a clear role, and monthly or quarterly review meetings ensure that issues are surfaced and resolved quickly rather than allowed to fester. This structure respects IOM's

existing authority and systems while clarifying where Carbon Clarity adds value and who is accountable for what.

Carbon Clarity succeeds because it aligns design choices with real-world constraints. Low literacy is addressed through visual and oral communication rather than text-heavy tools. Trust deficits are mitigated by working through existing institutions such as VSLAs and IOM's grievance mechanisms rather than parallel systems. Market complexity is managed through phased learning rather than one-off training, and financial sustainability is ensured by embedding the model into ongoing operations instead of relying on standalone technology or staffing. The sustainability model assumes no increase in per-site costs, ensuring that financial viability is driven by revenue diversification rather than cost escalation. We assume the Carbon Project Developer will provide unredacted access to pricing and cost data. This is a critical-path dependency; without this data, the $14,700 investment in transparency materials cannot achieve its goal of closing the information gap.

CONCLUSION AND NEXT STEPS

This report has presented a problem and a solution. The problem is that displaced communities in northern Mozambique generate climate value through their daily use of improved cookstoves, yet they cannot see, understand, or negotiate how that value is priced and shared. This information asymmetry leaves them vulnerable to exploitation and prevents them from making informed decisions about whether carbon finance actually serves their interests. The 2024 cookstove pilot, which delivered real immediate benefits but failed to deliver promised carbon income with any transparency, made this gap painfully clear.

Carbon Clarity responds to this problem with a practical, evidence-based system that does three things: it educates communities over time through trusted channels so they understand carbon concepts; it provides simple, visual tools so they can verify that pricing formulas are applied correctly; and it gives them a clear pathway to challenge unfairness and demand accountability. The system is deliberately low-tech and embedded in existing institutions, VSLAs, IOM's field presence, and established grievance mechanisms, so that it can be sustained after external funding ends. Testing with 50 household representatives across three sites showed that this approach is both understandable to people with limited formal education and acceptable to them; communities expressed preference for grievance routes that run through trusted local figures rather than distant hotlines, and they appreciated visual, icon-based tools over text-heavy documents.

If implemented at full scale across the nine IOM cookstove sites, Carbon Clarity aims to reach 2,770 households and deliver concrete outcomes by Year 3: 80 percent carbon literacy, 100 percent receipt-verified payments, 50 percent women in governance roles, 85 percent stove retention, and approximately 10,000 Meticais per site in community benefit funds. These outcomes are not isolated; together, they create a system where carbon finance becomes a reliable, understandable, accountable tool for both climate action and community resilience.

Expansion and Evolution

The success of the pilot creates opportunities for growth. Within Mozambique, the model can be adapted and replicated to other IOM cookstove sites and to different clean cooking technologies, biogas, improved charcoal, solar, as long as carbon finance is involved. The tools and training scripts are designed to be modular and translatable, meaning replication costs are marginal compared to the initial pilot. Beyond Mozambique, the same approach can be tested in other displacement and climate contexts across Africa, where similar gaps exist between community capacity and carbon market complexity. Discussions with IOM country offices in South Sudan, DRC, and Uganda suggest that there is strong institutional appetite for such a model.

Over time, Carbon Clarity can also evolve. As internet connectivity improves in resettlement areas, optional digital tools, such as SMS receipt confirmations or household-level payment tracking, can supplement low-tech boards and paper receipts without replacing them. If government frameworks for carbon finance mature in Mozambique, the transparency tools can be integrated into national registry and benefit-sharing requirements. The model can expand to address related challenges: youth entrepreneurship in briquette production, women's financial inclusion, or community adaptation to climate impacts. None of these expansions require abandoning the core principle, that communities must understand and control the terms on which their climate actions are valued and compensated.

The Path Forward

The next steps are clear. Funders should commit to supporting the full three-year pilot across nine sites, recognizing that transparency is not a cost but an investment in climate finance credibility. IOM Mozambique should formalize the partnership through an MoU and begin recruiting and training facilitators. The carbon project developer should sign a transparency commitment and prepare to share pricing and cost data with communities. Communities themselves should be invited to participate in co-design workshops and to hold their leaders and external actors accountable for delivering on promises. By taking these steps now, during the months between stove distribution and credit registration, all actors can ensure that when carbon payments finally arrive, communities will be ready to receive them fairly, transparently, and with confidence that the system works.

References

Ecosystem Marketplace (2022). *State of the Voluntary Carbon Markets* – Africa regional data.

IOM (2024). *Displacement Tracking Matrix – Mozambique* (Cabo Delgado, Nampula and Niassa).

IOM & UN Women (2022). *Gender and Equity Outcomes in Community Carbon Projects in Africa*.

OCHA (2024). *Mozambique Humanitarian Response Plan* – Displacement figures.

SEEP Network (2023). *Savings Groups Evidence and Learning Brief* – Sub-Saharan Africa VSLA data.

World Bank (2023). *Carbon Finance in Sub-Saharan Africa: Opportunities and Risks*

APPENDICES

APPENDIX A: SAMPLE VSLA TRAINING MODULES

APPENDIX B: SAMPLE PRICE BOARD DESIGN

WANACHO WA MBUZI - CHETI YA BEI (OUR STOVE MONEY - PRICE CERTIFICATE)

SOKO YA ULIMWENGU | | WORLD MARKET PRICE

[GLOBE ICON] $12.00 per tonne CO_{2e} This is what the world pays for carbon reductions

KUMI CHAKO - SHARES

YOUR SHARE

Community receives: 60% of market price $12.00 • 60% = $7.20 per tonne

KADIMU ZILIZOTOLEWA - EXPENSES

COSTS THAT COME OUT

[TRUCK ICON] Transport - $1.00

[WAREHOUSE ICON] Storage - $0.50

[CHECKMARK ICON] Verification - $0.30

SUBTOTAL COSTS - $1.80

MALIPIRO YAKO

WHAT YOU RECEIVE

[MONEY ICON] $7.20 - $1.80 = $5.40 per tonne

This is what households receive per tonne of carbon

SANDUKU LA KOMYUNITI

COMMUNITY BENEFIT FUND

[SCHOOL ICON] [W$0.40 per tonne goes to projects YOUR VSLA chooses:

ELL ICON] [CLINIC ICON]

Water • School • Health • Other community needs

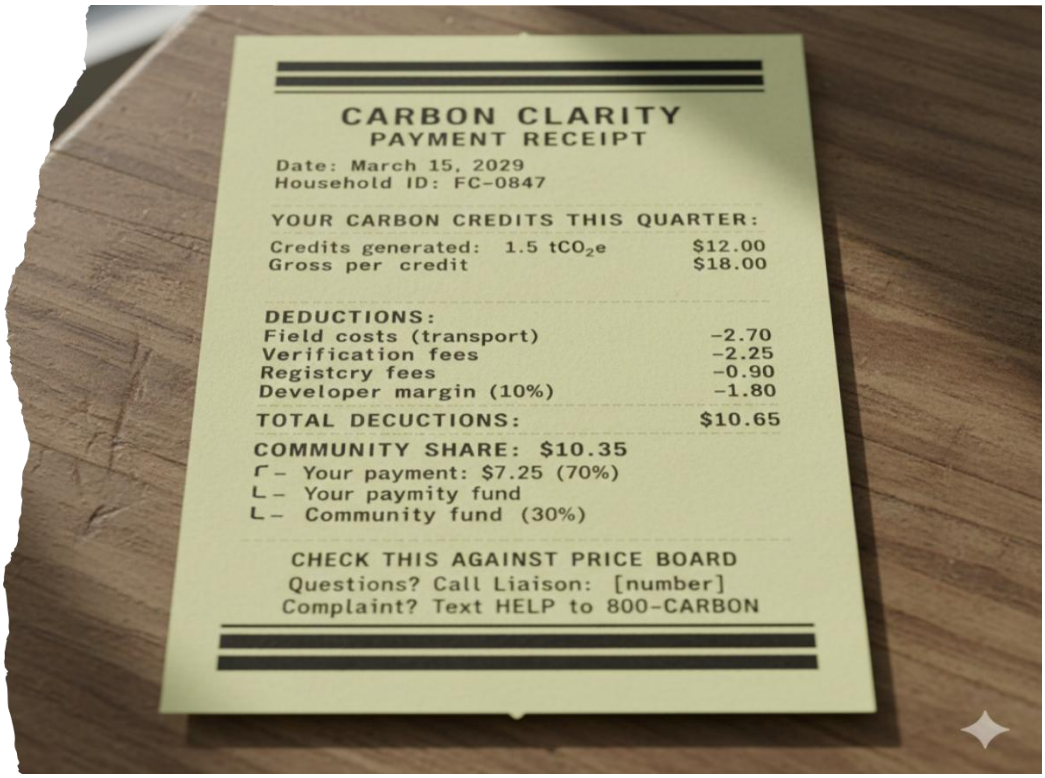
Last Updated: _____

Questions? Ask [VSLA LEADER NAME] or contact the Carbon Liaison

REPORT UNFAIR PRICING: Contact IOM Field Office within 14 days



APPENDIX C: ITEMIZED RECEIPT TEMPLATE



CARBON CLARITY
PAYMENT RECEIPT

Date: March 15, 2029
Household ID: FC-0847

YOUR CARBON CREDITS THIS QUARTER:

Credits generated: 1.5 tCO_2e	$12.00
Gross per credit	$18.00

DEDUCTIONS:

Field costs (transport)	-2.70
Verification fees	-2.25
Registcry fees	-0.90
Developer margin (10%)	-1.80
TOTAL DECUCTIONS:	$10.65

COMMUNITY SHARE: $10.35
- Your payment: $7.25 (70%)
- Your paymity fund
- Community fund (30%)

CHECK THIS AGAINST PRICE BOARD
Questions? Call Liaison: [number]
Complaint? Text HELP to 800-CARBON

APPENDIX D: CARBON OVERSIGHT COMMITTEE CHARTER

APPENDIX E: VSLA FACILITATOR JOB DESCRIPTION

APPENDIX F: TIMELINE & DELIVERABLES CHECKLIST

APPENDIX G: RISK REGISTER & MITIGATION STRATEGIES

APPENDIX H: EVIDENCE BASE & KEY ASSUMPTIONS

APPENDIX I: COMPARISON TO STANDARD PRACTICE

APPENDIX J: FREQUENTLY ASKED QUESTIONS (FAQ)

APPENDIX K : SUPPORTING RESEARCH AND DATA

APPENDIX L : SUCCESS STORY TEMPLATE

APPENDIX M: FEEDBACK INTEGRATION MATRIX

APPENDIX N : ADDENDUM

APPENDIX O: MEMORANDUM OF UNDERSTANDING

APPENDIX P: Data Protection Feels "Defensive"

APPENDIX Q: Comparative Case Study Analysis

APPENDIX R: Success Metric: Trust Recovery

APPENDIX S: SUSTAINABILITY STRATEGY

APPENDIX T: FINANCIAL CASE

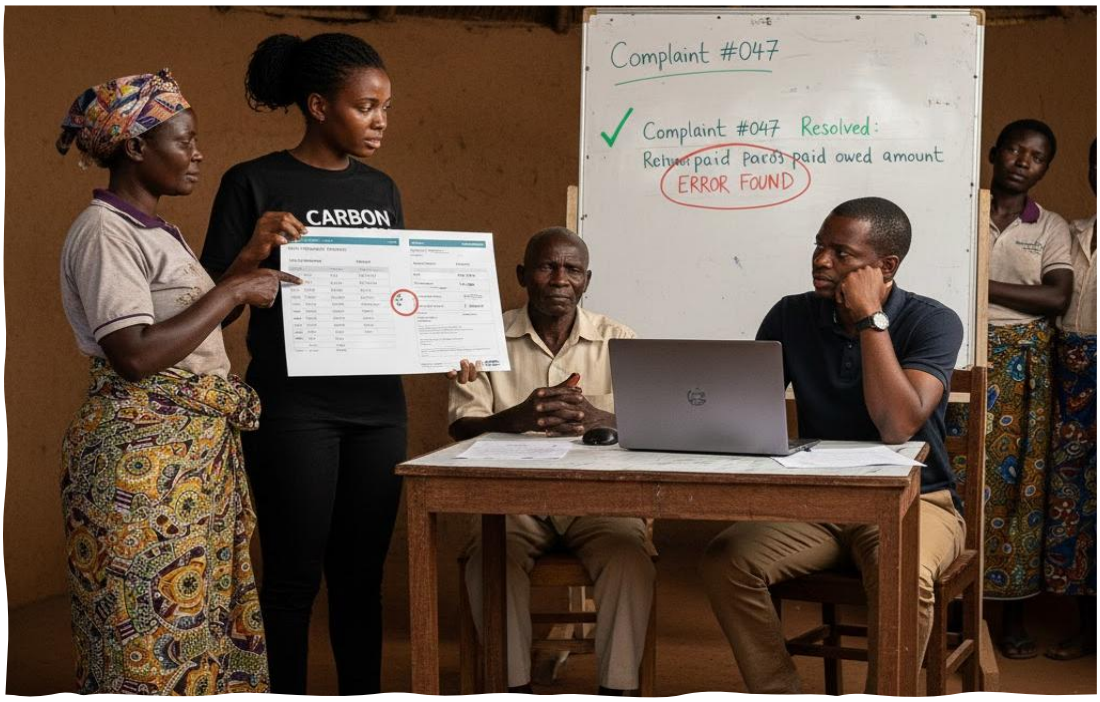
APPENDIX U: BUSINESS MODEL

Business Model Canvas: Carbon Clarity

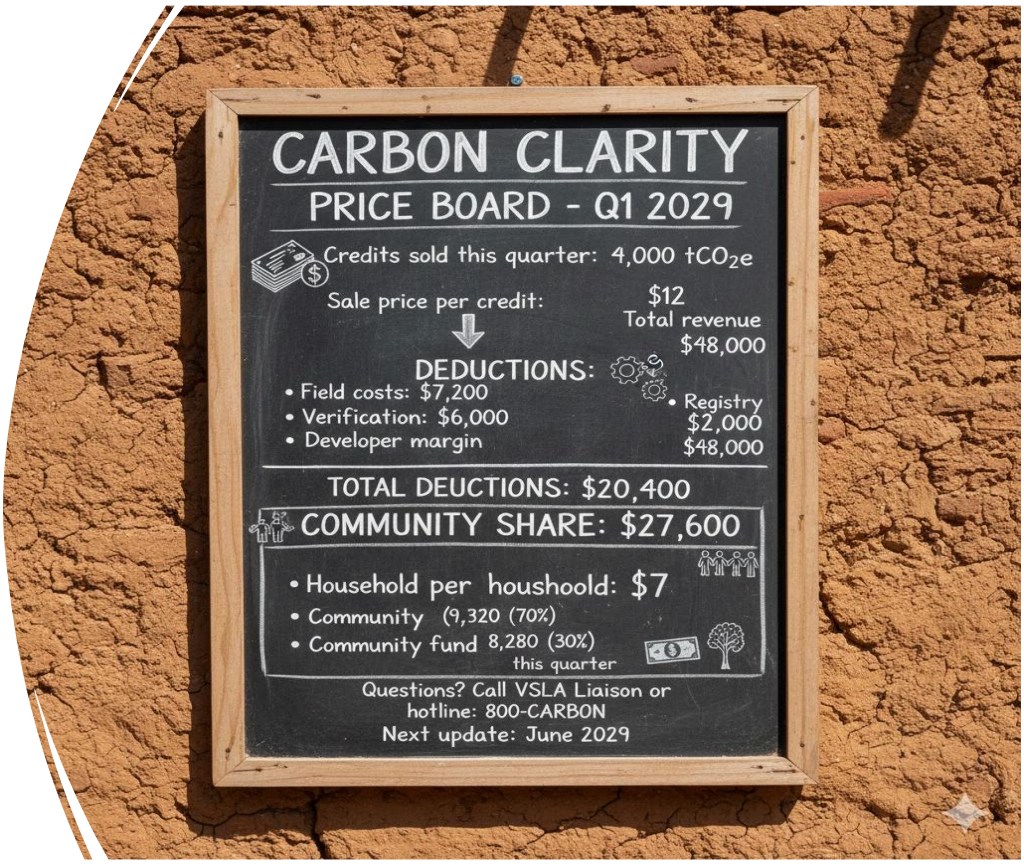
Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
IOM & Government Project Developers Standards Bodies VSLA & Mobile Money feedback VSLAs committees Government reporting Community Radio	◦ Credit aggregation ◦ Transparent price setting ◦ Household payments ◦ Benefit fund management ◦ Education & Hub training	For Communities: • Fair prices • Simple contracts • Clear receipts • Protection from exploitation For Buyers: • High-integrity • credits • Low risk	◦ Long-term trust via community committees ◦ regular feedback loops ◦ dedicated account managers for buyers ◦ transparent public reporting	Primary (IDP) ◦ VSLA (Village Savings and Loan Association) groups. Secondary ◦ Corporate carbon buyers and project developers. Tertiary ◦ Government bodies and international donors.
	Key Resources		**Channels**	
	◦ People: Staff, VSLA ◦ Systems: Encrypted data, MRV ◦ Physical: Office, price boards, ◦ Relationships: IOM, developers, ◦ Data encryption & tech ◦ Local office & physical price boards		◦ VSLA meetings ◦ Community gatherings ◦ Local offices ◦ Mobile money/SMS ◦ Community radio ◦ Price boards	

Cost Structure	Revenue Streams
Personnel: $63,000/year facilitators, liaisons Transport: $6,000/year tracking Materials: $3,000/year (guides, boards) Office/admin: $8,000/year communities Training: $6,000/year Technology: $3,000/year Tech	Carbon margin: 10-15% of sales (Year 3+) Service fees: $0.50-1.00/household/year Developer fees: $0.05-0.10 per tCO_2e rants: $60k/year (Year 1-2) Training/licensing fees (Year 4+)

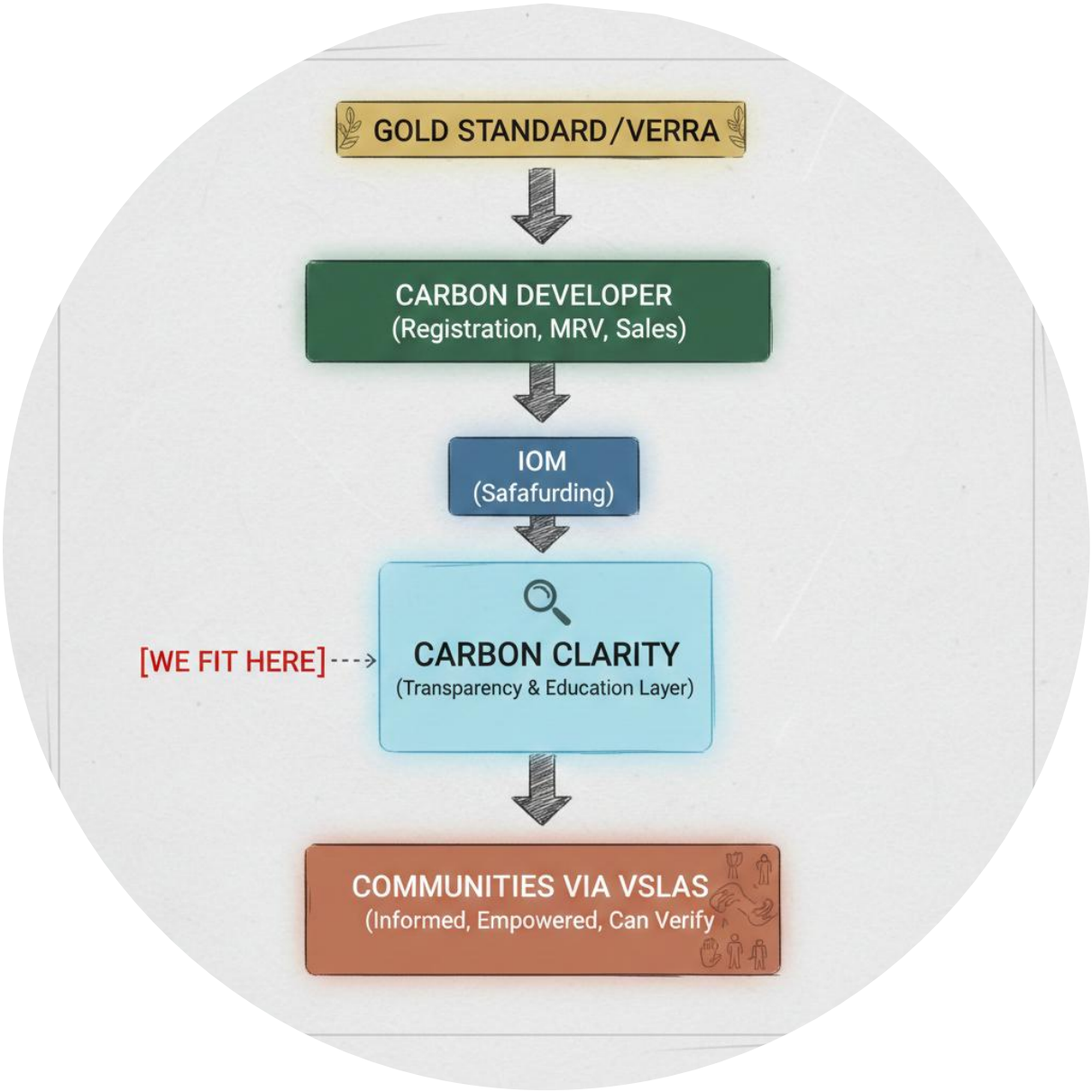
APPENDIX V: USER JOURNEY



APPENDIX W: PRICE BOARD



APPENDIX X: WHERE WE FIT IN THE SYSTEM



APPENDIX X: LINK TO EXCEL SHEET

APPENDIX Y: TECHNICAL & OPERATIONAL FEASIBILITY

APPENDIX Z: GENDER & INCLUSION